

Literature Review on Price Detection of Used Cars

1. Evaluation of Machine Learning Techniques for Used Car Price Prediction:

Author: Abdulla AlShared (Rochester Institute of Technology)

Overview: This research investigates the application of data mining and supervised learning to address information asymmetry in the used car market. The study emphasizes the economic significance of the automotive sector and the necessity for objective valuation tools to protect consumers from price manipulation. AlShared evaluates a range of machine learning paradigms—including supervised, unsupervised, and reinforcement learning—to determine their efficacy in predicting resale values. The paper highlights that accurate predictive models not only aid buyers but also provide financial institutions with more robust frameworks for assessing loan risks and collateral value.

2. Predicting Used Car Price Based on Machine Learning: A Comparative Study:

Author: Jiayi Lin (2024)

Overview: This study presents a comparative analysis of three prominent regression models: Linear Regression, Ridge Regression, and Random Forest. Utilizing the CarDekho dataset, the author focuses on the rigor of the data preprocessing phase, including outlier detection, missing value imputation, and feature encoding. The findings indicate that the Random Forest algorithm significantly outperforms traditional linear models in terms of predictive accuracy and handling non-linear relationships between vehicle attributes (e.g., age, mileage, and fuel type). The research concludes that while linear models offer interpretability, ensemble methods are superior for capturing the complexities of market fluctuations.

3. Used Car Price Prediction Based on LightGBM and XGBoost:

Source: ACM Digital Library (2024)

Overview: This paper explores the transition from traditional regression to advanced gradient-boosting frameworks. Specifically, it compares the performance of LightGBM (Light

Gradient Boosting Machine) and XGBoost (Extreme Gradient Boosting) in the context of high-dimensional automotive data. The study demonstrates that LightGBM offers superior computational efficiency and training speed compared to XGBoost, without a significant loss in accuracy. This research is particularly relevant for real-time valuation platforms that require high-velocity data processing. It emphasizes the critical role of hyperparameter optimization in achieving high R-squared values and low Mean Absolute Error (MAE).

4. Predicting the Price of Used Cars using Machine Learning:

Source: IEEE Xplore (2023)

Overview: Published in the proceedings of the International Conference on Evolutionary Algorithms and Soft Computing Techniques, this paper focuses on the development of a statistical model using a Kaggle-sourced dataset. The research primarily utilizes Linear Regression to establish a baseline for price forecasting. The authors provide a detailed breakdown of the correlation between various vehicle features and market price. With an R-squared value of 0.829, the study validates the effectiveness of basic regression techniques for standard datasets. It serves as a benchmark for how specific variables, such as "Kilometers Driven" and "Year of Manufacture," act as the strongest predictors of vehicle depreciation.

5. Used Car Price Prediction Using Machine Learning Techniques:

Source: Academia.edu (Rosemol Thomas, 2022)

Overview: This research addresses the global surge in used car sales driven by the increasing cost of new vehicles. The paper compares Linear Regression and Lasso Regression to mitigate the issue of overfitting often found in high-dimensional datasets. The study highlights the advantages of Lasso Regression in performing feature selection by shrinking the coefficients of less important variables to zero. The results show that Lasso Regression (87.09% accuracy) provided a more reliable prediction than standard Linear Regression (83.65%). The paper also includes the development of a user interface (UI), demonstrating the practical application of these models for end-user car valuation.

6. Car Price Prediction using Machine Learning (2024):

In this paper the author presents a machine learning based approach for car price using several algorithms. The study includes data processing, how to handle missing values, normalizing dataset, removing outliers. The essential steps are shown here to reduce inconsistencies of model accuracy.

The features in this paper used such as:

- Car brand and model
- Fuel Type (petrol/diesel/else)
- Manufacture date and year
- The capacity of engine
- Type of Transmission

Algorithms used in this paper include: Linear Regression, Decision Tree, Random Forest. Among all these Random Forest performed really well, when it comes to handling nonlinear relationships and reducing overfitting. In this paper it shows that car age, mileage have a strong impact on price and also analyzed other features. On the other hand, this paper has a limited discussion on real-time data application and it does not explore deep learning models. So, these are the limiting facts.

7. Used Car Price Prediction Using Machine Learning (2023):

This paper is mainly about predicting the price of used cars, which is actually more complex than others due to the condition and depreciation facts. The author has displayed that the price of used cars is influenced by many dynamic factors such as market demand, previous ownership.

The used dataset in this paper:

- Linear Regression
- Decision tree
- Random forest

Also, it includes (EDA) to identify correlations, visualize price distribution and trends. The features here are mileage, age, fuel type and seller type. The ultimate result shows that Linear regression isn't worth it for complex patterns; Decision tree improves the accuracy. Random Tree Forest has the best achievement among all. The limitation is that the dataset is small and it does not include external information of the market or trends.

8. Pre-Owned Car Price Prediction: A Machine Learning Approach (2025):

Here in this paper, it builds a simple interpretable model using Linear Regression for predicting resale value of cars. The priority here is the relation between features and price rather than accuracy only.

The methodology here:

- Data Cleaning
- Feature Encoding
- Exploratory Data Analysis for identifying trends.

- Model Building

The results here show that Linear Regression provides reasonable prediction but still struggles with complex nonlinear relationships. Limiting factor is low accuracy and cannot capture feature interactions.

Reference List:

- **AlShared, A.** (2020). *Evaluation of machine learning techniques for used car price prediction and valuation* [Master's thesis, Rochester Institute of Technology]. RIT Digital Library. <https://repository.rit.edu/theses/10660/>
- **Lin, J.** (2024). Predicting used car price based on machine learning: A comparative study. In *Proceedings of the 1st International Conference on Big Data and Artificial Intelligence (BDAI 2024)* (pp. 351–356). SciTePress. <https://doi.org/10.5220/0013270500003923>
- **Liu, Y., & Zhang, H.** (2024). Used car price prediction based on LightGBM and XGBoost. *ACM Digital Library*. <https://doi.org/10.1145/3674029.3674032>
- **Sahoo, S., & Panda, A.** (2023). Predicting the price of used cars using machine learning. In *2023 International Conference on Evolutionary Algorithms and Soft Computing Techniques (EASCT)* (pp. 1–5). IEEE. <https://doi.org/10.1109/EASCT59475.2023.10393486>
- **Thomas, R.** (2022). *Used car price prediction using machine learning techniques*. Academia.edu. https://www.academia.edu/98178262/Used_Car_Price_Prediction_Using_Machine_Learning_Techniques
- Ahmad, M. F., et al. (2024). Car Price Prediction using Machine Learning. <https://share.google/daBZdTS1YeMocXLYS>
- Raghav, S. S., et al. (2023). Used Car Price Prediction Using Machine Learning. <https://share.google/gS7Jp8K76uuuMK65K>
- Marnholkar, T. (2025). Pre-Owned Car Price Prediction: A Machine Learning Approach. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5189395